

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: Create Notes app and build for web deploy it on GitHub pages also create release APK and appbundle.	
Experiment No:- 10	Date:- 20-11-2024	Enrollment No:- 92200133030	

Objective:- Create Notes app and build for web deploy it on GitHub pages also create release APK and appbundle.

Code :-

➤ Lib

- **models**

- **notes.dart**

```
class Note {
  String? title;
  String? description;
  String? id;
  Note({this.title, this.description, this.id});
}
```

- **services**

- **firebase_services.dart**

```
import 'package:cloud_firestore/cloud_firestore.dart';
import '../models/notes.dart';

class FirebaseServices {
  var db = FirebaseFirestore.instance;
  addNote(String title, String description) {
    // Create a new note with a title and description
    final note = <String, dynamic>{"title": title, "description": description};

    // Add a new document with a generated ID
    db.collection("notes").add(note).then((DocumentReference doc) =>
      print('DocumentSnapshot added with ID: ${doc.id}'));
  }

  Stream<QuerySnapshot> fetchNotes() {
    Stream<QuerySnapshot> collectionStream =
      FirebaseFirestore.instance.collection('notes').snapshots();
    return collectionStream;
  }

  // Stream documentStream =
  FirebaseFirestore.instance.collection('users').doc('ABC123').snapshots();

  updateNote(Note note) {
    return db
      .collection("notes")
```

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```

        .doc(note.id)
        .update({'title': note.title, 'description': note.description})
        .then((value) => print("Note Updated"))
        .catchError((error) => print("Failed to update Note: $error"));
    }

    deleteNote(id) {
        return db
            .collection("notes")
            .doc(id)
            .delete()
            .then((value) => print("User Deleted"))
            .catchError((error) => print("Failed to delete user: $error"));
    }
}

```

- **screen**

- **notes_details_screen.dart**

```

import 'package:flutter/material.dart';

import '../models/notes.dart';
import '../services/firebase_services.dart';

class NotesDetailScreen extends StatefulWidget {
    final Note? note;
    NotesDetailScreen({this.note});
    @override
    State<NotesDetailScreen> createState() => _NotesDetailScreenState();
}

class _NotesDetailScreenState extends State<NotesDetailScreen> {
    final _formKey = GlobalKey<FormState>();
    late String _title;
    late String _content;

    @override
    void initState() {
        super.initState();
        if (widget.note != null) {
            _title = widget.note!.title!;
            _content = widget.note!.description!;
        } else {
            _title = '';
            _content = '';
        }
    }

    void _saveNote() {

```

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```

    if (_formKey.currentState!.validate()) {
      _formKey.currentState!.save();
      final note = Note(
        id: widget.note?.id,
        title: _title,
        description: _content,
      );
      //todo code to store on firestore
      FirebaseServices().addNote(_title, _content);
      Navigator.pop(context);
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(widget.note == null ? 'New Note' : 'Edit Note'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16.0),
        child: Form(
          key: _formKey,
          child: Column(
            children: [
              TextFormField(
                initialValue: _title,
                decoration: InputDecoration(labelText: 'Title'),
                validator: (value) {
                  if (value == null || value.isEmpty) {
                    return 'Please enter a title';
                  }
                  return null;
                },
                onSave: (value) {
                  _title = value!;
                },
              ),
              SizedBox(height: 16.0),
              TextFormField(
                initialValue: _content,
                decoration: InputDecoration(labelText: 'Content'),
                validator: (value) {
                  if (value == null || value.isEmpty) {
                    return 'Please enter content';
                  }
                  return null;
                },
                onSave: (value) {
                  _content = value!;
                },
              ),
            ],
          ),
        ),
      ),
    );
  }
}

```

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```

        SizedBox(height: 16.0),
        ElevatedButton(
          onPressed: _saveNote,
          child: Text('Save'),
        ),
      ],
    ),
  ),
);
}
}

```

○ notes _list_screen.dart

```

import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter/material.dart';
import 'package:notes_app/services/firebase_services.dart';

import '../models/notes.dart';
import 'notes_details_screen.dart';

class NotesListScreen extends StatelessWidget {
  const NotesListScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Notes'),
      ),
      body: StreamBuilder<QuerySnapshot>(
        stream: FirebaseServices().fetchNotes(),
        builder: (BuildContext context, AsyncSnapshot<QuerySnapshot> snapshot) {
          if (snapshot.hasError) {
            return Text('Something went wrong');
          }

          if (snapshot.connectionState == ConnectionState.waiting) {
            return Center(child: CircularProgressIndicator());
          }

          if (!snapshot.hasData ||
              snapshot.data == null ||
              snapshot.data!.docs.isEmpty) {
            return Text("No notes found");
          }
        }
      )
    );
  }
}

```

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```

return ListView(
  children: snapshot.data!.docs.map((DocumentSnapshot document) {
    Map<String, dynamic>? data =
      document.data() as Map<String, dynamic>;

    if (data == null) {
      return ListTile(
        title: Text('No Data'),
      );
    }

    // Safely handle null for title and description
    String title = data['title'] ?? 'Untitled';
    String description = data['description'] ?? 'No description';

    Note note = Note(
      id: document.id,
      title: title,
      description: description,
    );

    return GestureDetector(
      onDoubleTap: () {
        FirebaseServices().deleteNote(document.id);
      },
      onTap: () {
        Navigator.of(context).push(MaterialPageRoute(
          builder: (context) => NotesDetailScreen(note: note)));
      },
      child: ListTile(
        title: Text(note.title!),
        subtitle: Text(note.description!),
      ),
    );
  }).toList(),
);
),
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Navigator.of(context).push(
      MaterialPageRoute(builder: (context) => NotesDetailScreen()));
  },
  child: Icon(Icons.add),
),
);
}
}

```

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Output:-

